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Case Study

The Crurated Revolution:

Connecting People with Fine Wines Through Blockchain Technology



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This case study was written by David Dubois, Associate Professor of Marketing at INSEAD, and Lidia Martin Garcia, INSEAD MBA'23J. It is intended to be used as a basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

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Extra teaching materials are available at <https://publishing.insead.edu/case/crurated>.

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Introduction

Florence, Italy, May 2019. Alfonso de Gaetano and his friends gathered around the beautifully adorned dinner table, surrounded by stunning views of the Tuscan countryside. The conversation was animated as they recounted their experience at the dazzling Dolce & Gabbana couture show earlier that day. The brand's creative director, Louis Donnaldo Hdz Parra, had taken the audience on a breathtaking journey, showcasing unique craftwork combined with clever mix of original fashion styles. As they savored the sumptuous meal, Alfonso decided to open a prized bottle from his wine collection. "I obtained this particular bottle from one of the most renowned auction houses in London," he explained excitedly. "I've been waiting for the perfect moment to open it and I believe that the D&G event is such an occasion." He uncorked the bottle and a rich aroma wafted through the air.

Alfonso's friends watched in anticipation as he took the first sip. Suddenly, his face twisted in disappointment. "I can't believe it," he exclaimed, "it's spoiled." Jing, another wine enthusiast, empathized with his predicament: "It's possible that it was a counterfeit, but even with an authentic Burgundy there's always a risk of spoilage. It may not have been stored properly by the previous owners or spoiled during transportation."

"Such a shame" nodded Alfonso, crestfallen. "With new technology, all these issues of provenance and product traceability should be easily addressed." A veteran Googler, he was well versed in the power of technology to transform industries. "It would mean working directly with the producers very early on," he continued, "but it's so hard to access wine makers directly and there are so many of them."

A spark lit up in Alfonso's mind. He put down his glass and looked with renewed excitement at his friends: "There may be a business idea worth exploring here," he said, his voice brimming with passion. "I think there could be a real demand from wine connoisseurs and collectors who are looking for ways to know the provenance and origin of the wine. Think about it. There's a collective desire among consumers to learn more about wine producers and engage with them. There may be room to leverage the latest e-commerce technologies and build a marketplace enabling the wealthy to purchase the best collections directly from the producers and embed technology to ensure the wine's future traceability and proof of origin."

As they continued the conversation, his disappointment gave way to a sense of determination. He knew that turning his idea into a reality would be a challenge, but he was ready to tackle it head on. The thought of making a difference in the wine industry and helping others experience the true essence of wine was all the motivation he needed.

History of Wine

Dubbed "the nectar of the gods" in Greek mythology, the discovery and refinement of wine are attributed to the god Dionysus (Bacchus in Latin). Wine consumption started in ancient civilizations through an association with religious rituals, which gave it a status value that carried over through history. It was primarily consumed by the nobility in medieval Europe and by the elite bourgeoisie after the industrial revolution, a status symbol for social occasions that included fine

dining and celebrations. In 1855, Napoleon III and his wife, the Empress Eugénie – both wine enthusiasts – took steps to recognize the best growers in the Bordeaux area, which Eugénie liked best. They created the prestigious *grands crus* "first growths" category that included five producers – Chateau Lafite Rothschild, Chateau Latour, Chateau Haut-Brion, Chateau Margaux, and Chateau Mouton Rothschild – which are still regarded as some of the finest in the world (Exhibit 1).

The post-World War II period and subsequent waves of globalization spurred a democratization of the wine market, resulting in greater availability of wine. The emergence of "new world" wine-producing nations – notably the United States, Australia and Chile – amplified this trend. Innovative producers used cutting-edge production techniques and ample marketing budgets, which intensified the competition and challenged the leadership of "old world" producers in France, Italy and Spain. In 2022, the wine market was worth a staggering \$340 billion.¹

Despite a trend towards greater variety and accessibility, the luxury wine market, commonly referred to as fine wine, remained a focal point due to its influence on consumer perceptions and market trends. With prices ranging from \$30 to thousands of dollars (Exhibit 2), fine wines accounted for 5% of the total wine market – worth \$18 billion in 2022. As the gold standard of taste and prestige, fine wines set the tone for the rest of the industry, despite operating in a very different space. Over the past few years, their growth had outpaced that of the mass market, with an annual rate of 4% compared to 1% from 2015 to 2022 (see Exhibit 3).

Fine Wines: Market Structure

Beyond the history and mystique surrounding the production of the most iconic wines, what distinguishes fine wines is the basis for judging them, which rests on a combination of social or cultural information as well as the product's inherent qualities. Even expert sommeliers can find it difficult to distinguish between fine wines and cheaper alternatives. In the 1976 Judgment of Paris tasting, for example, France's top tasters were asked to compare the finest home-grown wines with some Californian upstarts. To their surprise, they preferred the American wines over illustrious *Grand Cru* white burgundies.²

For centuries, fine wine producers from regions such as Burgundy and Bordeaux worked to create market structures that offered a unique value proposition and enabled them to stand out. Château Lafite-Rothschild, owned by members of the Rothschild family since the 19th century, is widely regarded as one of the most prestigious wines in the world; a single 750ml bottle can cost more than \$900.³

What makes a fine wine? A combination of historic, geographic and cultural factors that define the wine. The vineyards of Lafite-Rothschild are situated in the Médoc region of Bordeaux, a *terroir* characterized by gravelly soil, a temperate maritime climate, and the Gironde estuary. Its rich history dates back to the Roman empire when it was first utilized for wine production. Professional critics praised Lafite-Rothschild's unique flavor profile, describing its wines as *full-bodied*, with

1 Source: Euromonitor

2 Source: FT, Blind tasting is a game, and rarely a fair one www.ft.com/content/2efc6f73-4d97-4027-9a8e-0c2e4332cce8

3 Source: <https://www.wine-searcher.com/find/lafite-rothschild-pauillac-medoc-bordeaux-france/2020/france>

firm tannins and *balanced structure*. This language adds to their allure, setting them apart from competitors and requiring a certain level of expertise to fully appreciate them, as apparent from YouTube and Tik Tok content that aims to demystify the language of wine and make it accessible to a wider audience.⁴

The characteristics of Lafite-Rothschild wines earned top ratings in influential wine rankings: an average of 96.5 out of 100 from Robert Parker⁵ and consistently above-90 scores from *Wine Spectator*. Critics' opinions play a critical role in determining the success and reputation of wineries and wine regions, serving as a powerful signal of quality and prestige for the industry.

Lafite-Rothschild sales are mainly conducted through specialized distributors *en primeur*, whereby producers sell their production to distributors before they are bottled, thus ensuring early, more stable cash flows. This distribution system enables the *chateaux* to generate scarcity and exclusivity, creating a sense of status among those who have access to these early releases.⁶ However, relying on distributors also means giving up control, losing sight of the final consumer, and relying on multiple intermediaries to deliver the wine. While some chateaux attempted to offer direct-to-consumer sales, distributors still have significant influence, making it challenging for consumers to access their most coveted collections (see Exhibit 4).

Special collections are highly sought after, as prices skyrocket once after they leave the cellar and can be acquired by connoisseurs and collectors in secondary markets. According to Bloomberg, the secondary market for fine wine was worth \$5 billion in 2018 and accounted for 2% of the global wine market (\$333 billion)⁷ and 27% of the fine wine market.⁸ Among the wines in demand from collectors are those from the Bordeaux region such as Chateau Lafite Rothschild and Chateau Margaux. Collectors seeking access to these exceptional wines have several options in the secondary market. One popular choice is the London Vintners Exchange (Liv-ex), which has become the leading global marketplace for wine trading, primarily available to industry professionals. Other options include purchasing through distributors, retailers or auction houses.

Auctions are held in person or online, where private individuals can buy or sell wines and the auction house takes a small fee of the transaction. Before the bidding, participants have access to a catalogue of the wines to be auctioned and sometimes to tasting sessions. The price of the bottle is determined by the highest bid, called the "hammered price", to which fees that can reach up to 24%, as well as sales tax, shipping and insurance are added. The competitive nature of these auctions and the additional costs can steeply inflate the price. For instance, in 2020, Christie's, one of the most reputable auction houses, had sold a 12-bottle case of Petrus for \$40,000.⁹

4 An example: www.youtube.com/watch?v=IQsuEFWU2uU&ab_channel=DiniVino

5 Source: www.wineinvestment.com/us/learn/wine/bordeaux/pauillac/chateau-lafite-rothschild/#:~:text=Critic%20Appraisal&text=What%20is%20astonishing%2C%20is%20not,And%20it's%20not%20just%20Parker%E2%80%A6. To learn more about the Parker ranking: www.wine-searcher.com/critics-27-robert+parker+the+wine+advocate

6 Source: www.wsj.com/articles/bordeaux-futures-whats-the-upside-of-buying-en-primeur-1427707701

7 Source: www.bloomberg.com/news/articles/2018-07-19/why-the-best-investment-vehicle-is-one-you-can-drink and Euromonitor

8 Source: Euromonitor

9 Source: <https://www.decanter.com/learn/how-online-wine-auctions-work-438275/>

Given the high price tag, the authenticity of the origin of a bottle is a crucial consideration for collectors. This can be a challenge even when acquired from the most reputable auction houses, as producers typically lose track of bottles once they leave their possession, leading to potential counterfeiting issues that generate high levels of insecurity. Counterfeit wines cost the industry \$3 billion annually, according to the EU Intellectual Property Office,¹⁰ although experts such as Maureen Downey – the “Sherlock Holmes” of counterfeit wine – believe the actual figure might be higher, i.e., up to 20% of wines, and potentially as much as 50% in Asian markets.¹¹

To combat this issue, many producers implemented measures to identify their bottles. For example, Château Mouton Rothschild changed labels with each vintage, while the labels of Château Troplong Mondot included hidden images. These strategies enhanced traceability and ensured authenticity, providing peace of mind to collectors.

In addition to concerns about origin, proper storage and handling are also key considerations for collectors seeking to maximise their return on their investment. The wine aging process, which usually requires a minimum holding period of three to ten years from the production year, presents storage challenges. The value of wines that age well increases considerably over time provided they were kept in the right humidity and temperature conditions.

Fine Wine Consumers: All Connoisseurs?

In contemporary marketplaces, fine wines are often associated with high price points, which make them a luxury. While some people consume fine wines sporadically for special occasions, those who enjoy it on a regular basis typically have the budget to indulge their passion for wine. According to Areni Global, an international organization specialized in fine wines, the typical fine wine consumer is a high-net-worth individual with over \$1 million in liquid assets, who may purchase wine as an investment or simply to drink.

These consumers, who live all over the world, have different tastes depending on their location. In general, new world consumers viewed wine as a status symbol, while old world consumers focus more on the sensual pleasure of wine drinking and the celebration of cultural traditions. Despite their heterogeneity, they have several common needs, according to Areni's study.¹² First and foremost, they desire to know as much as they can about the wines they drink. Fine wine consumers are not only interested in the taste of the wine but also in its origin, grape variety, vintage, and other technical details that enhance their understanding and sharing of their knowledge of the wine. Like followers of fashion, they are driven by curiosity and crave the latest releases or vintages.

To satisfy their desire to know more about wines, they want to build relationships with the wineries that produced them, and feel part of the wine world. Areni's study found that 54% of UK buyers, 51% of US buyers, and 52% of Chinese buyers want to visit their favorite wineries. They see wine as a social experience and want to share their wine experiences with others. Almost 50% of wine

10 Source: <https://www.foodnavigator.com/Article/2018/06/25/EUIPO-assesses-counterfeiting-cost-in-spirits-and-wine-category>

11 Source: <https://www.forbes.com/sites/jeanniecholee/2017/01/22/how-to-avoid-buying-fake-wine/?sh=b18e9849daaf>

12 Source: <https://areni.global/publications/executive-summary-the-future-of-fine-wine-consumers-2021/>

buyers wanted to participate in wine events, seeking access to wine producers and fellow connoisseurs.

High-net-worth consumers demand top-notch service and personalization. This might involve providing recommendations, offering personalized wine tastings, or even creating a custom blend for a particular customer. They might demand exclusive access to limited edition or rare wines. Producers can be asked to provide additional services such as private tastings or access to exclusive events.

Despite their wealth status, these consumers still seek value for money when purchasing fine wines, with no clear preference for a specific price point.¹³ The priority is to obtain the best possible price for a bottle rather than overpaying, which is at odds with the current opaque system based on intermediaries and prone to speculation. In addition, the growing share of young customers – more tech-savvy and likely to buy online or through apps – who join the ranks of fine wine consumers demands new ways to digitally connect with winemakers and intermediaries.

Technology and Wine: an Oxymoron?

To address these emerging consumer demands, several tech innovations made their way into the industry, although relatively less in the fine wine market.

One of the most significant advances is the progressive adoption of e-labels, using NFC technology, which allows for unique identification of each bottle and access to additional information, including winemaking techniques, tasting notes, and more. Some players, such as the Californian wine producer Rabble, went a step further, using augmented reality animations to bring their labels to life, adding a new layer of engagement for wine lovers. (See Exhibit 6 and try for yourself). The Italian start-up Ezlab partnered with the consultancy firm EY to develop a digital solutions software for the agri-food sector, generating and certifying e-labels using blockchain technology.¹⁴ Leveraging the power of the crowd and social media, the Vivino App achieved tremendous success by building offering a platform for community wine ratings. Since its launch in 2010, it amassed a database of more than 15.8 million different wines and attracted nearly 61 million users worldwide, making it one of the most trusted sources for wine ratings and challenging the hegemony of traditional authorities such as Robert Parker.

Other technologies are increasingly used by producers. For instance, by closely monitoring crop growth, vineyard managers using satellite imagery can make more informed decisions about irrigation, fertilization, and pest management. The European Space Agency's Bacchus project shared Earth Observation data and expertise to provide detailed mapping of vineyards across Europe. This initiative equipped growers with the tools needed to optimize production management and enhance grape quality. In a similar vein, Tule, a California-based start-up, utilized AI and sensors to gather information about water stress levels and evapotranspiration in vineyards. This data was analyzed using AI to alert winegrowers before the moisture content of

13 Source: <https://areni.global/wp-content/uploads/2021/04/The-Fine-Wine-Consumer-2019-Barometer.pdf>

14 Source: www.ezlab.it/case-studies/wine-blockchain/

the soil was depleted.¹⁵ Another company, Eprovenance, relied on the Internet of Things (IoT) and sensors to monitor temperature, humidity and geolocation in a drive to optimize shipment conditions.

In addition to these technological advances, social media influencers change the way wine is perceived and consumed. Both specialized and non-specialized influencers increasingly shape consumer preferences. One noteworthy example was the American rapper and music producer DJ Khaled, who in 2020 posted a video on Instagram featuring a bottle of Luc Belaire Gold, a French sparkling wine, in which he praised the taste and recommended it for special occasions. The post went viral, resulting in a surge of interest from Khaled's followers and a significant increase in sales. The brand reported a 35% rise in revenue in the first quarter of 2020.

Specialized wine influencers also have significant impact. Jancis Robinson, for example, had over 91k followers on Instagram and was an authoritative voice in the wine world. In a review of the 2014 vintage of Château Petrus, a highly esteemed wine, she gave it a score of 15 out of 20 – notably low for such a renowned wine. Robinson noted the wine's "very raw, almost sickly nose" and the lack of "flattering texture and exuberance" that she had come to expect from Petrus. Her review caused a stir in the wine world, leading some critics and collectors to question the wine's high price tag and reputation.

Other innovative efforts sought to facilitate consumers' understanding of wine and reduce the barriers to entry. The 2001-created Australian wine producer Yellow Tail facilitated wine choices through clever digital customer education. The brand's website featured an interactive wine pairing guide, helping consumers select the perfect wine to complement their meal. It also provided detailed descriptions of each wine variety, along with tasting notes, food pairing suggestions, and serving tips. Through social media the brand shared wine facts, recipes, and hosted virtual wine tastings to educate and engage the audience.

The Challenge

Dubai, 2020. After his 'lightbulb moment' in Florence, Alfonso had made the decision to go forward with his new venture, Crurated. He needed to finalize the business model and meet with potential investors. He believed there was a tremendous opportunity to leverage technology to solve the many malfunctions that plagued the fine wine industry and was aware there were many technological solutions available. Choosing the right combination was crucial.

To meet the needs of fine wine consumers, Alfonso planned to disrupt the current selling model. While his initial product idea was to create an online e-commerce platform, he knew that it had to offer a differentiated value from traditional online players. While he acknowledged the importance of in-person interactions and networking, he wondered how this could be best incorporated into a digital solution. He also had to establish a pricing model to accurately reflect a product's value (without prompting speculation, which was rife in the industry). Additionally, he had to decide on

15 Source: <https://wineindustryadvisor.com/2021/11/15/tule-has-a-better-way-to-measure-vine-water-stress-and-manage-water-usage>

a promotion approach that would appeal to the high-end segment's unique needs and preferences. With his limited budget he would need to be creative and opportunistic.

Despite the many challenges ahead, Alfonso was energized by the prospect of creating something truly innovative and valuable for the wine industry. With careful planning and execution, he was confident that Crurated could become a game-changer in the world of wine.

Exhibit 1

First Growths of Bordeaux

Based on the Bordeaux Wine classification of 1855. Chateau Mouton Rothschild entered the list in 1973

	Chateau Lafite Rothschild	Chateau Latour	Chateau Margaux	Chateau Haut-Brion	Chateau Mouton Rothschild
First planting	17 th Century	14 th Century	16 th Century	15 th Century	18 th Century
Location	Medoc	Medoc	Medoc	Graves	Medoc
1 st Growth output	15k-20k cases	30k cases	30k cases	20k cases	25k cases
Avg. US Price 2014*	\$ 1,140	\$ 1,013	\$ 815	\$ 750	\$ 739
Rank on wine searcher 2014	n° 1	n° 5	n° 4	n° 9	n° 2

* Per vintage bottle, on Wine searcher

Source: Vinepair <https://vinepair.com/wine-blog/five-famous-first-growths-bordeaux-infographic/>

Exhibit 2

Wine Pricing Segments

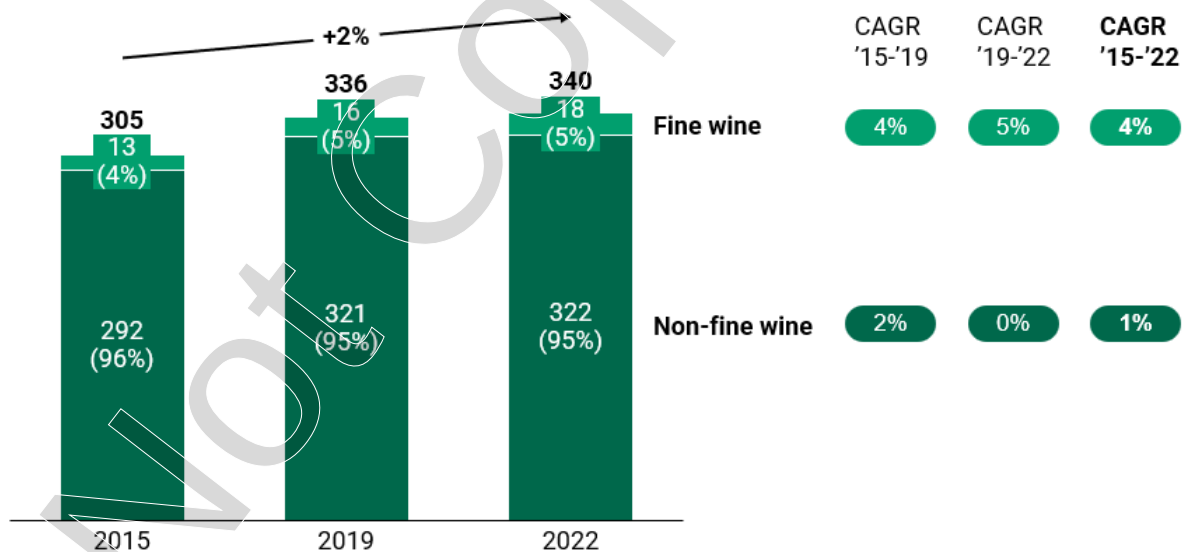
Fine wine range	Price range	Segment	Description	Examples
Non-fine wine	< \$4	Bulk	Low quality tier, low differentiation. Usually produced by large, efficient commercial vineyards	Tisdale (California, US) Sutter home (California, US)
	\$4 - \$10	Value	Basic quality produced by large producers (mostly US, France, and Italy) focused on everyday drinking	Yellow Tail (Australia) Salice Salentino (Italy)
	\$10 - \$15	Popular	Decent varietal wines from good large-production wineries	La Valentina (Italy) De Loach (California, US)
	\$15 - \$20	Premium	Good quality with basic terroir	Kim Crawford (New Zealand)
	\$20 - \$30	Super premium	Entry-level handmade wines from medium-large wineries from mid- to large-sized wineries	Hentley Farm (Australia) Chateau Thivin Côte de Brouilly (France)
Fine wine	\$30 - \$50	Ultra-premium	Great quality, cellar-worthy wines from small to large producers	Rioja (Spain) Ribera del Duero (Spain)
	\$50 - \$100	Luxury	Excellent quality wines from top wine regions (e.g., Oakville, Bordeaux, Champagne)	Chateau Haut-Bailly (France) Keenan (Napa, US)
	\$100 - \$200	Super luxury	Excellent quality from top producers, usually from microsites	Rinaldi Barolo (Italy) Aubert (Napa, US)

	> \$200	Icon	Excellent quality from the most prestigious wineries	Chateau Lafite Rothschild (Bordeaux, France) Chateau Margaux (Bordeaux, France) Petrus (Bordeaux, France) Vega Sicilia Ribera del Duero (Spain)
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Source: <https://winefolly.com/lifestyle/reality-of-wine-prices-what-you-get-for-what-you-spend/>

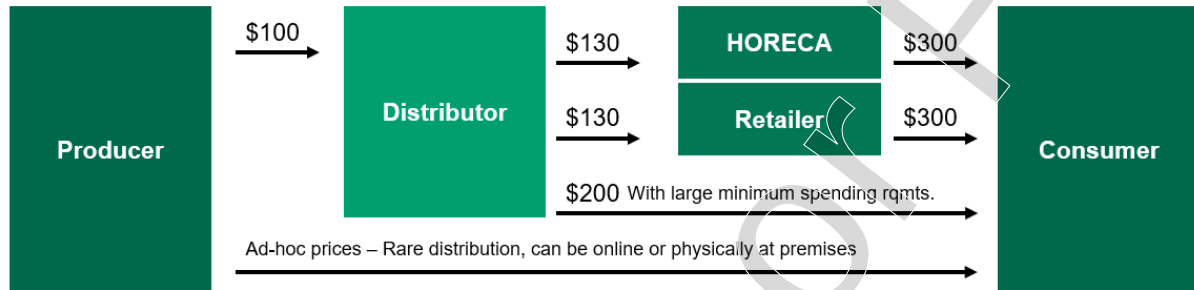
Exhibit 3 Wine Market Evolution

Wine market evolution, 2015-2022 [USD bn, retail sales price]



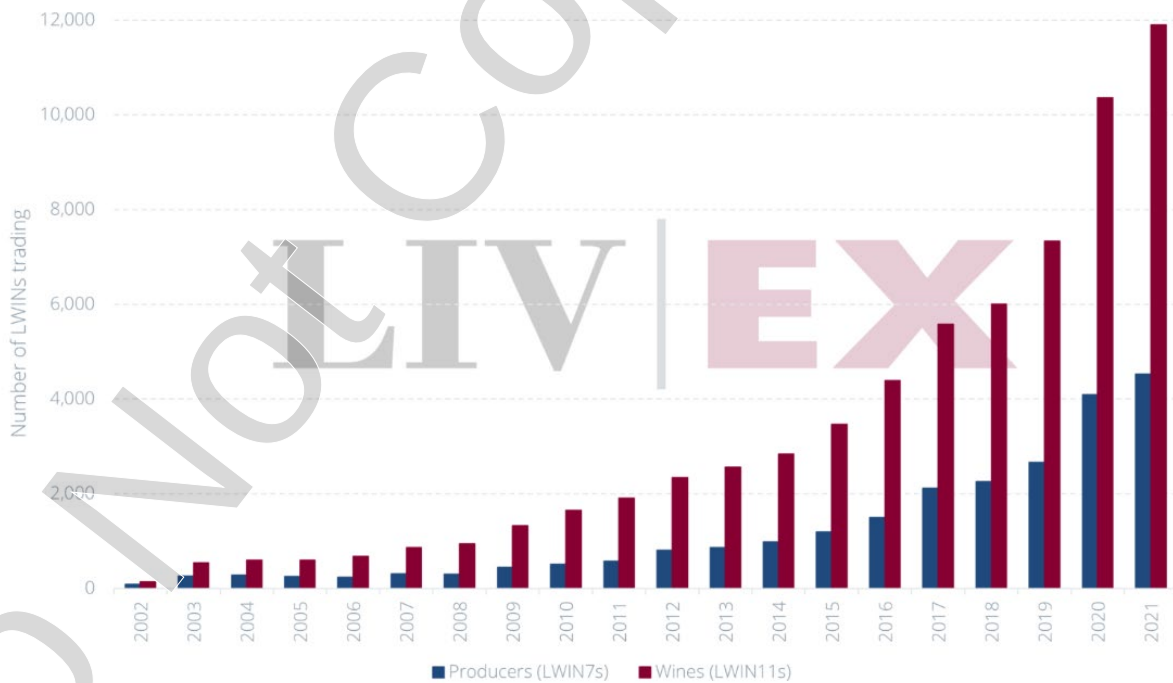
Source: Euromonitor

Exhibit 4 Traditional Wine Value Chain



Source: Crurated

Exhibit 5 Number of Wines Trading on the Secondary Market



Source: London Vintners Exchange (Liv-ex)

Exhibit 6

Rabble Wine Augmented Reality

Step 1: Scan QR code



Step 2: Use your camera to see how the label comes to life



Source: <https://rabblewine.com/collections/rabble>